

Eg:- Arrays, lists, stacks

Q. Findout the history of Towers of Hanoi

→ It is a famous mathematical puzzle invented by Edward Lucas in 1883.

In a temple in India, monks were given 3 rods and 64 golden disks stacked with 1 rod, with largest at bottom and smallest at the top.

→ The monks' task was to move all 64 disk to another rod, following these rules.

1. Move only one disk at a time.

2. Only top disk can be removed

3. A disk larger can never be placed on a smaller disk

→ The legend says that when the monks finish moving all 64 disks, the world will end

→ Minimum number of moves for 64 disks is $2^{64} - 1 = 18,446,744,073,709,551,615$ moves

Even if one move were made every second, it would take 584 billion years to finish - far longer than current age of universe.

Q. Today from the morning 7:00 am to 4:30 pm list out various data structures you observe.

1. Tooth brush holder - array
2. Soap, shampoo, face wash - array
3. Wardrobe clothes - stack (LIFO)
4. Breakfast - queue (FIFO)
5. Bus - queue (FIFO)
6. Seating in bus - array
7. Bus routed - VZM to MVGR to Lendi to Raghuto Anits
8. Google maps - linked list
9. Persons at stops (departing)
 - FIFO
 - LIFO